



Automating Your Architecture

AWS Academy Cloud
Architecting



Introduction

Automating Your Architecture

Module objectives

This module prepares you to do the following:

- Recognize when to use architecture automation and why.
- Identify how to use infrastructure as code (IaC) as a strategy for provisioning and managing cloud resources.
- Identify how to model, create, and manage a collection of AWS resources by using AWS CloudFormation.
- Identify how to use AWS Quick Start CloudFormation templates to set up an architecture.
- Identify uses of Amazon Q Developer.
- Use the AWS Well-Architected Framework principles when designing automation strategies.

Module overview

Presentation sections

- Reasons to automate
- Using infrastructure as code
- Customizing with CloudFormation
- Using AWS Quick Starts
- Customizing with AmazonQ Developer
- Applying AWS Well-Architected Framework principles

Demos

- Analyzing an AWS CloudFormation Template
- AWS CloudFormation Resources
- Reviewing an AWS CloudFormation Template
- Using the AWS CloudFormation Console

Knowledge checks

- 10-question knowledge check
- Sample exam question

Hands-on labs in this module

Guided lab

- Automating Infrastructure Deployment with AWS CloudFormation

Challenge (Café) lab

- Automating Infrastructure Deployment



As a cloud architect designing:



- I need to use automation to reduce the risks of manual processes when building architectures and maximize the benefit from reproducible environments.
- I need to deploy my architectures by using infrastructure as code so that I can rapidly and consistently deploy environments and changes.
- I need to consider how my automation design decisions impact performance and cost so that I can optimize the value of the architecture to the business.

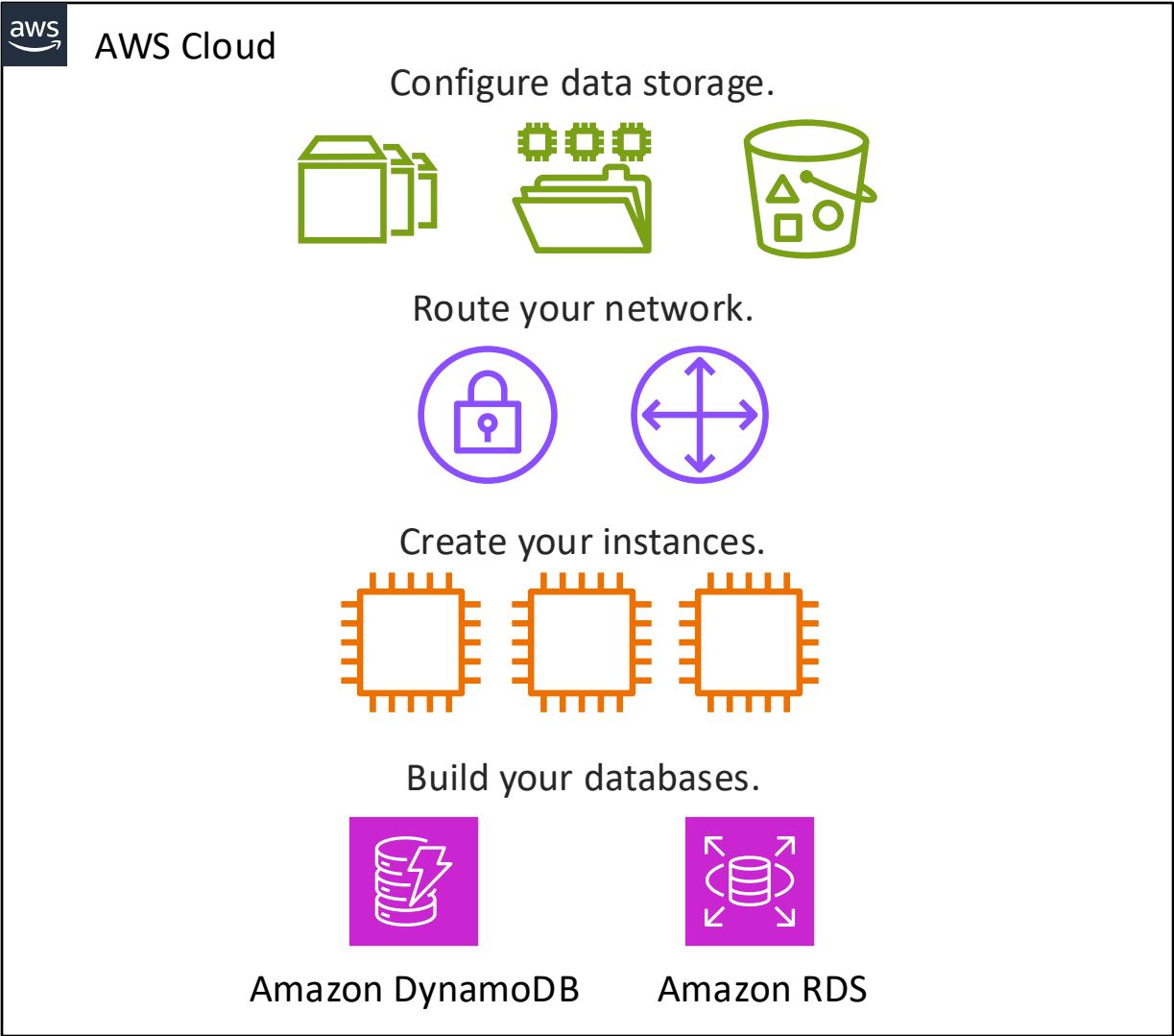
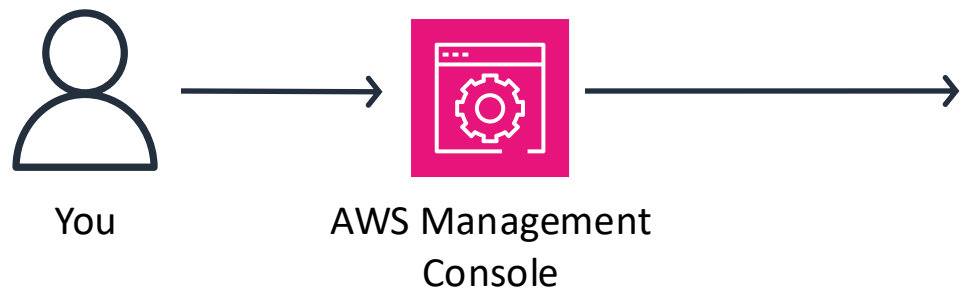


Reasons to automate

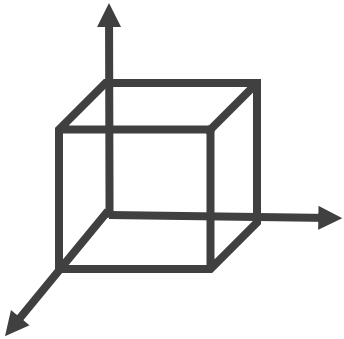
Automating Your Architecture

Process without automation

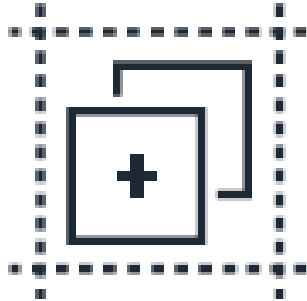
Long manual process to build an architecture



Risks from manual processes



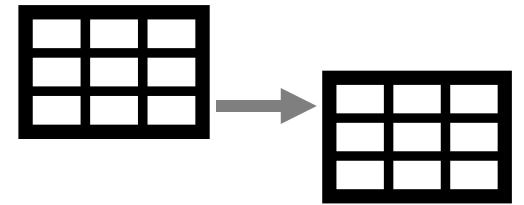
Does not support
repeatability at scale



Does not have version
control



Lacks audit trails



Has inconsistent
configurations

Benefits of automation



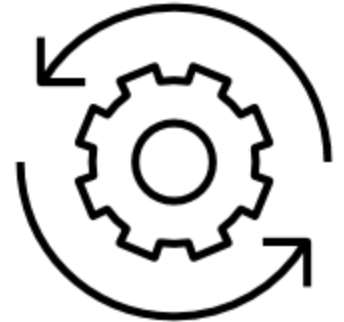
Reduces manual
intervention or access



Allows for reproducible
environments



Improves productivity



Automates testing
and scaling

Key takeaways: Reasons to automate



- Manual processes are error prone, unreliable, and inadequate to support an agile business.
- Manual processes create risks to applications and environments.
- Automation helps you eliminate manual process and build rapidly.



Using infrastructure as code

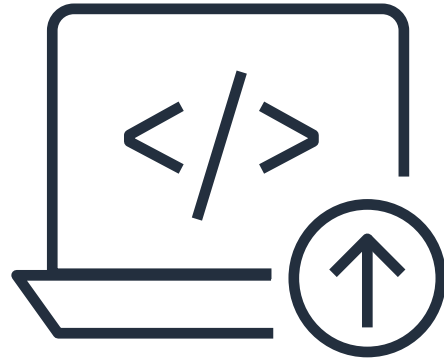
Automating Your Architecture

IaC overview

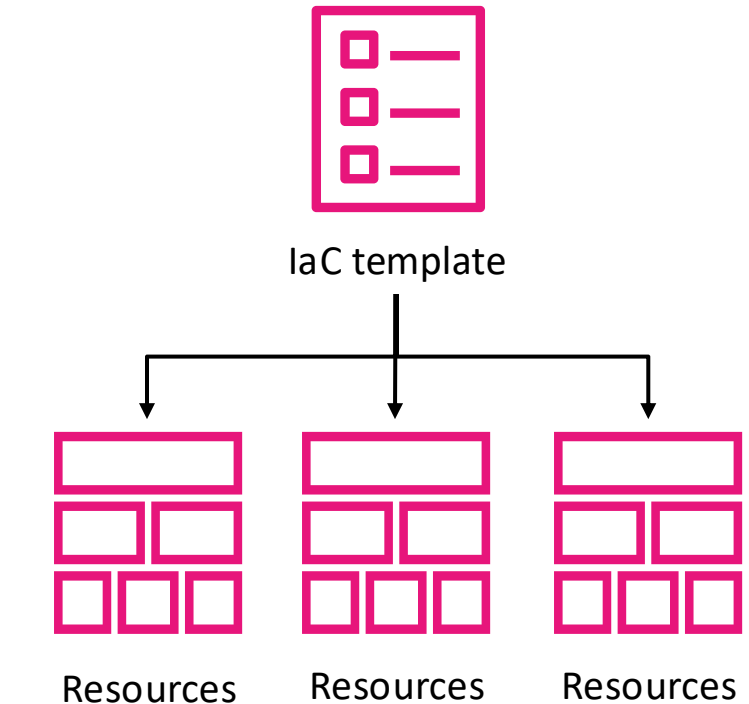
IaC is the process of writing a template that provisions and manages your cloud resources.



Human readable



Machine consumable



With IaC, you can replicate, redeploy, and repurpose infrastructure.

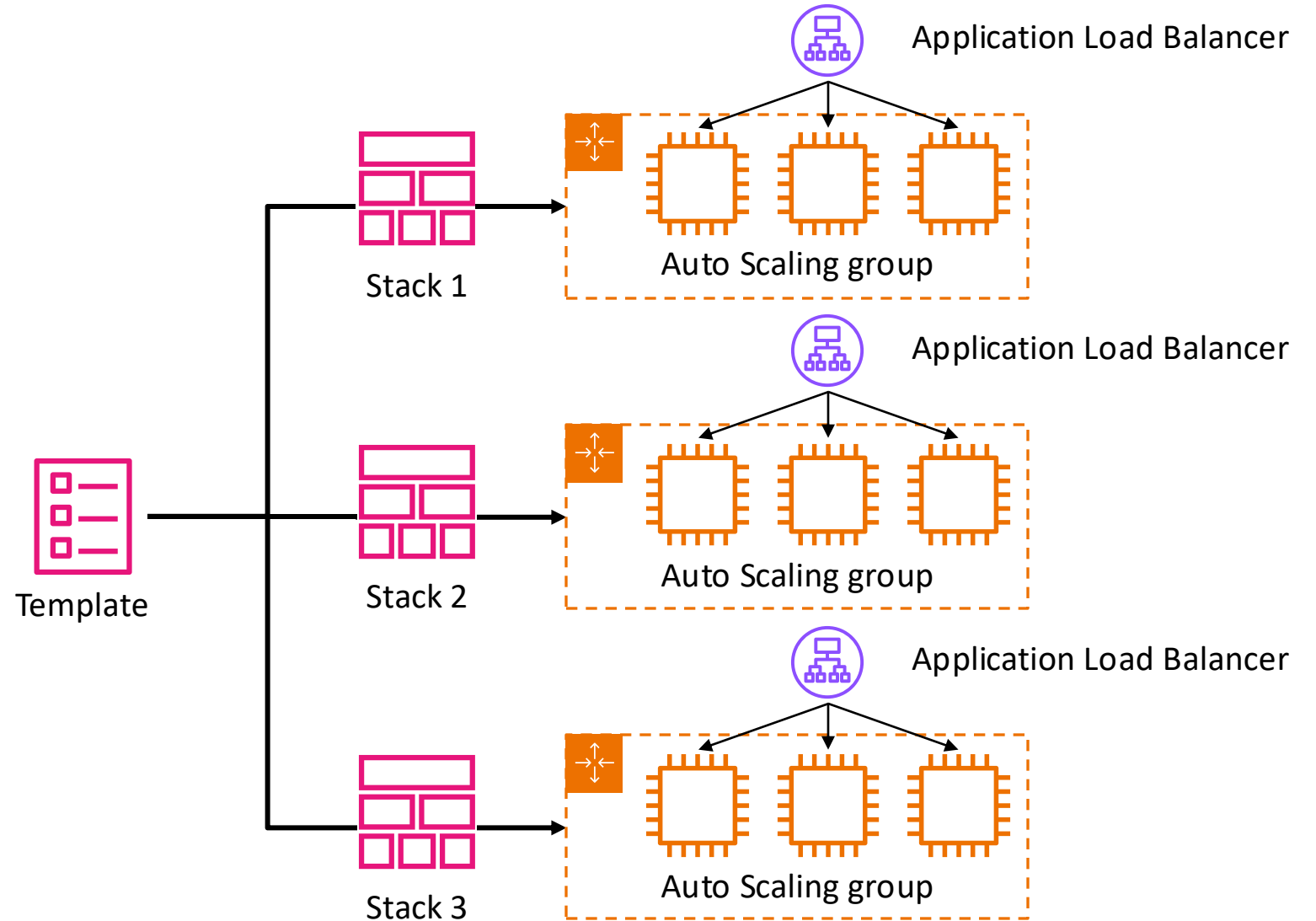
IaC benefits

Rapidly deploy complex environments with configuration consistency.

Propagate a change to all stacks by modifying the template.

Clean up by deleting the stack, which deletes the resources created.

The key benefits are reusability, repeatability, and maintainability.



CloudFormation



CloudFormation

- Provides a simplified way to model, create, and manage a collection of AWS resources
 - A collection of resources is called a CloudFormation stack.
 - There is no extra charge (pay for only the resources that you create).
- Can create, update, and delete stacks
- Enables orderly and predictable provisioning and updating of resources
- Enables version control of AWS resource deployments

AWS IaC services that use CloudFormation



AWS Elastic Beanstalk

A fully managed service that automatically launches an AWS environment with your uploaded application code



AWS Quick Starts

Automated reference architectures based on CloudFormation templates



AWS Serverless Application Model (AWS SAM)

An extension of CloudFormation with shorthand syntax for building common serverless infrastructure



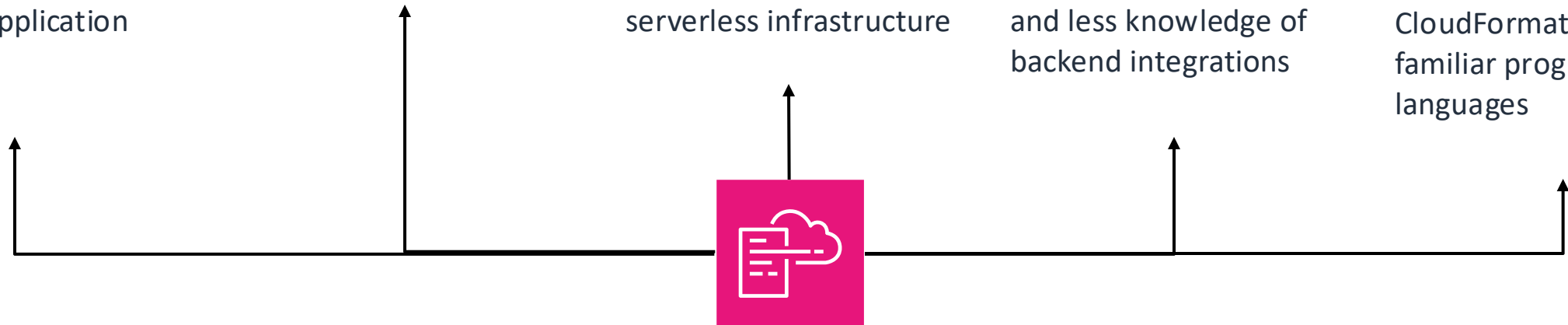
AWS Amplify

A development framework for full stack web and mobile apps that requires less coding and less knowledge of backend integrations



AWS Cloud Development Kit (AWS CDK)

An open source development framework to model and provision resources through CloudFormation by using familiar programming languages



AWS CloudFormation

Key takeaways: Using infrastructure as code



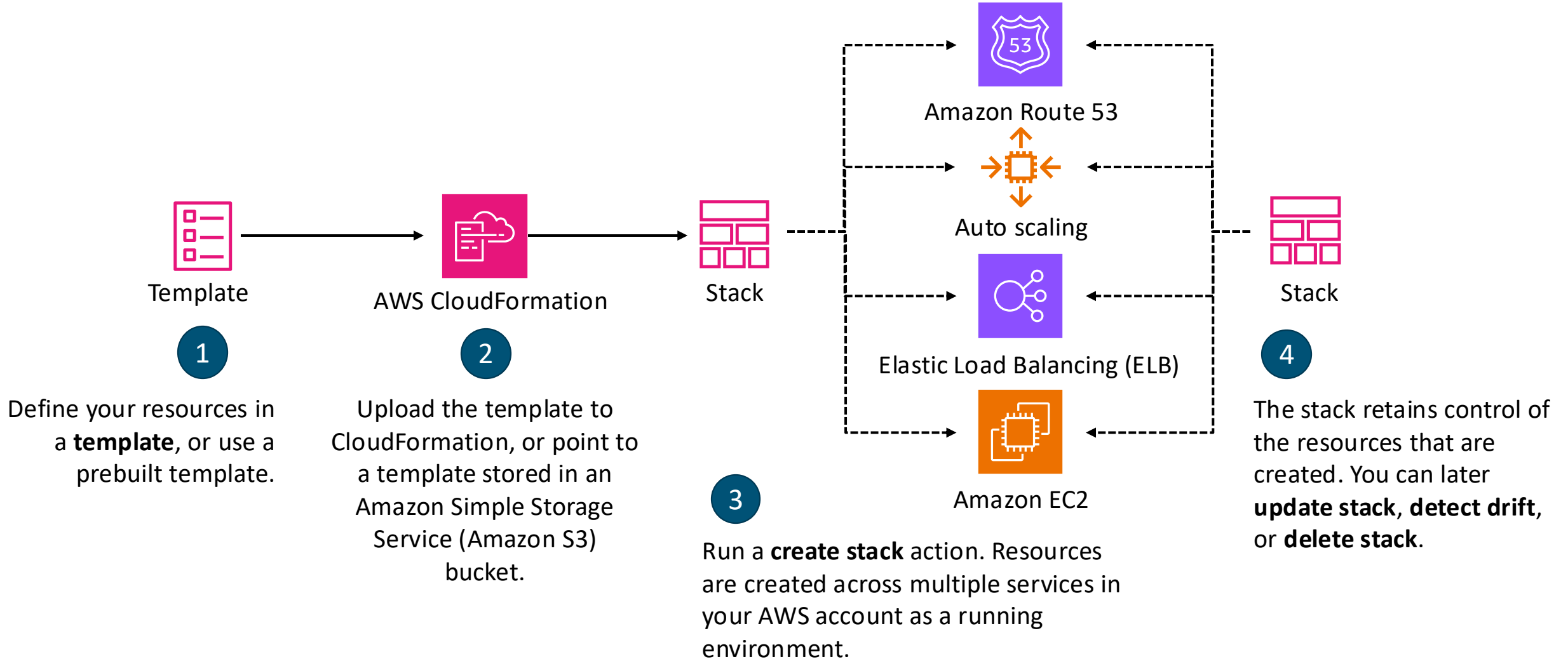
- IaC is the process of provisioning and managing your cloud resources by writing a template file.
- IaC rapidly deploys complex environments, and you can build or update the same complex environments repeatedly.
- Choose an IaC solution based on your use case, relative balance of convenience and control, and the skills of your team.
- CloudFormation is an IaC service to help you create, update, and delete services and architectures.



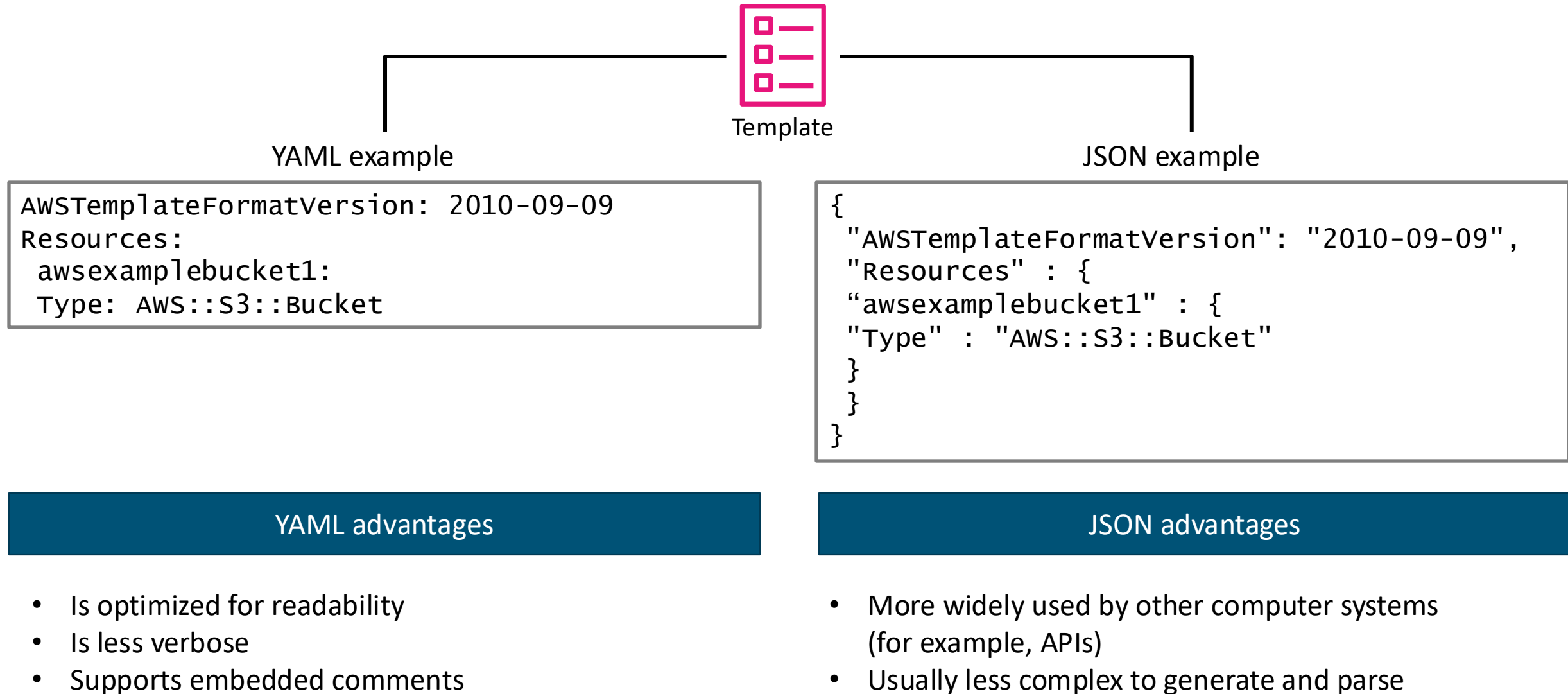
Customizing with CloudFormation

Automating Your Architecture

How CloudFormation works



CloudFormation template syntax



Anatomy of a CloudFormation template

- When you write a CloudFormation template you have a choice of sections to include based on your workload.
- The only required section is **Resources**. Additional sections are optional.
- This example shows a YAML-formatted template fragment in the suggested order of sections.

AWSTemplateFormatVersion: *"version date"*

Description:

String

Metadata:

template metadata

Parameters:

set of parameters

Rules:

set of rules

Mappings:

set of mappings

Conditions:

set of conditions

Transform:

set of transforms

Resources:

set of resources

Outputs:

set of outputs

Resources template: Create an EC2 instance

```
{
  "Resources": {
    "Ec2Instance": {
      "Type": "AWS::EC2::Instance",
      "Properties": {
        "ImageId": "ami-9d23aeea",
        "InstanceType": "m3.medium",
        "KeyName": {"Ref": "KeyPair"}
      }
    },
    "Outputs": {
      "InstanceId": {
        "Description": "InstanceId",
        "Value": {"Ref": "Ec2Instance"}
      }
    }
  }
}
```

← **Resources** define what needs to be created in the AWS account (all you need).

← **Outputs** specify the values returned after the stack is created.

AWS CloudFormation Designer (Designer)

The screenshot displays the AWS CloudFormation Designer interface. At the top, a toolbar (1) contains icons for file operations and a 'Close' button. On the left, a 'Resources types pane' (2) lists various AWS services such as ACMPCA, APS, AccessAnalyzer, AmazonMQ, Amplify, AmplifyUIBuilder, ApiGateway, ApiGatewayV2, AppConfig, and AppFlow. The central 'Canvas pane' (3) shows a visual diagram of a 'CustomerGateway' resource connected to an 'ElasticLoadBalancer' and a 'WebServer AutoScaling' resource. On the right side of the canvas, a 'Fit-to-window button' (4) and 'Full screen and Split screen buttons' (5) are visible. The bottom section features an 'Integrated JSON and YAML editor pane' (6) showing the template code for 'CustomerGateway' in YAML format. To the right of the editor is a 'Messages pane' (7) displaying a success message: '11/22/2023, 1:38:19 PM - Successfully converted the template to YAML.' The interface also includes a 'Choose template language' dropdown set to 'YAML' and tabs for 'Components' and 'Template' at the bottom left.

1
Toolbar

2
Resources types pane

3
Canvas pane

4
Fit-to-window button

5
Full screen and Split screen buttons

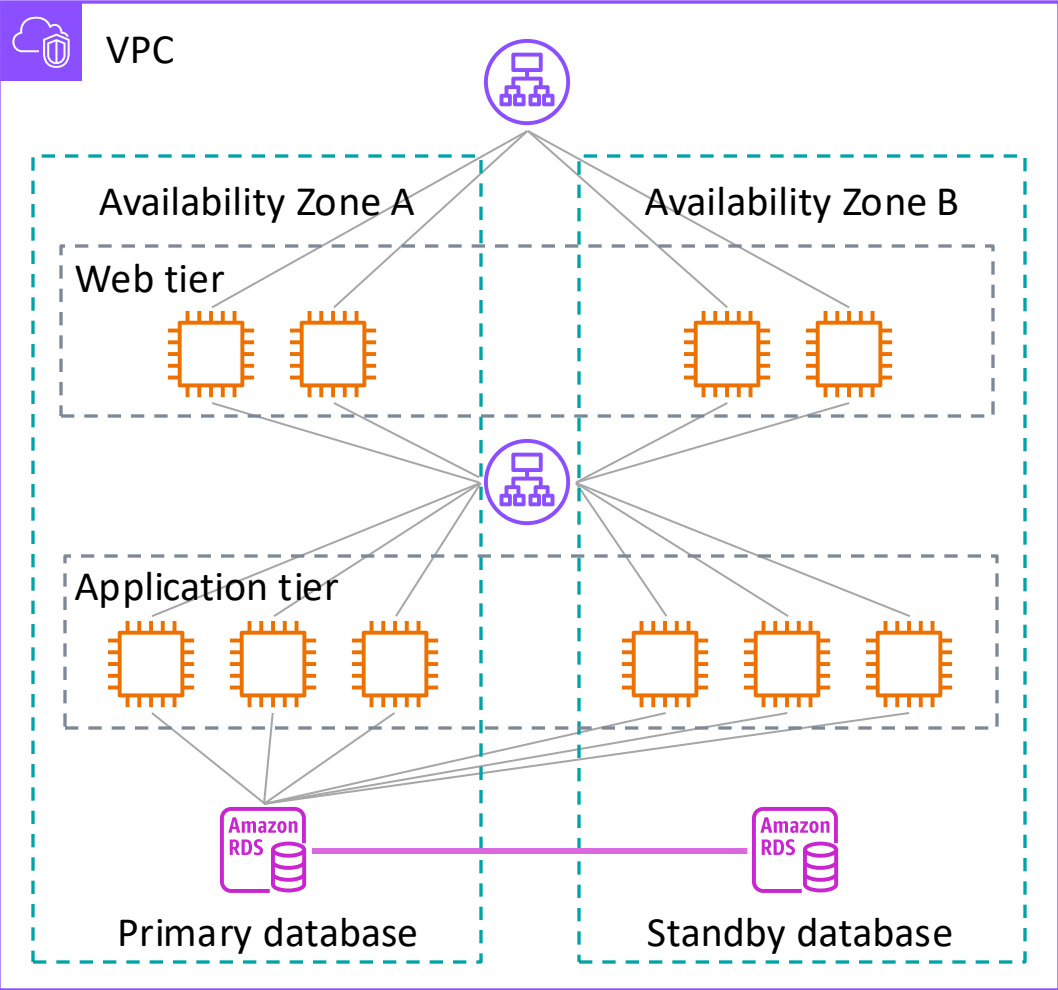
6
Integrated JSON and YAML editor pane

7
Messages pane

```
CustomerGateway
1 AWSTemplateFormatVersion: 2010-09-09
2 Metadata:
3   'AWS::CloudFormation::Designer':
4     17c5dc34-2ee3-45f5-aaa4-040cde9f136f:
5       size:
6         width: 60
7         height: 60
8       position:
9         x: 260
10        'y': 160
11       z: 0
12       embeds: []
13       a37cb14f-bcd7-4edc-b9c1-a8e7c11de311:
```

Using conditions

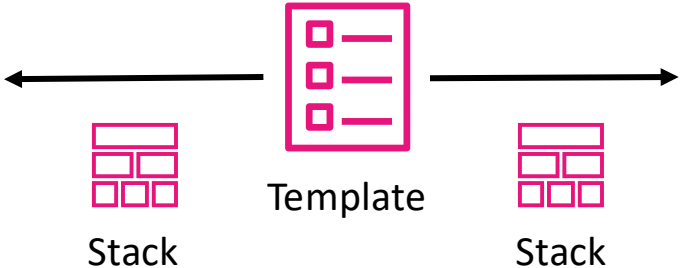
Production



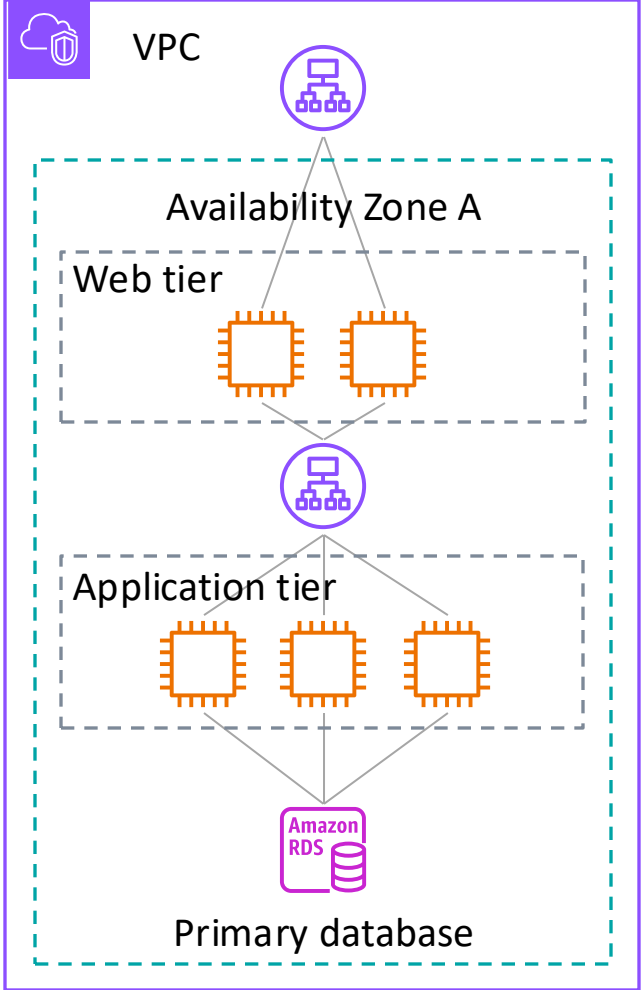
Conditions provisioned:

- Two Availability Zones in the production environment
- One Availability Zone in the development environment

This was done using the same template.

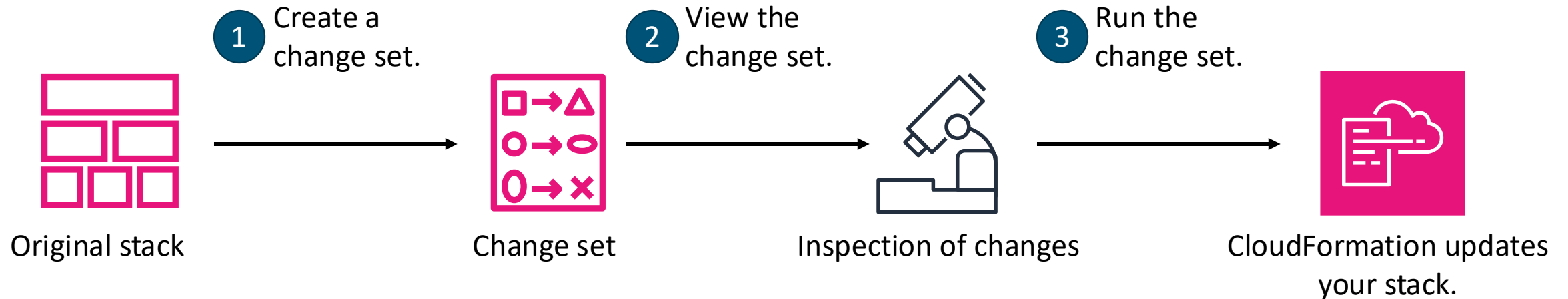


Development



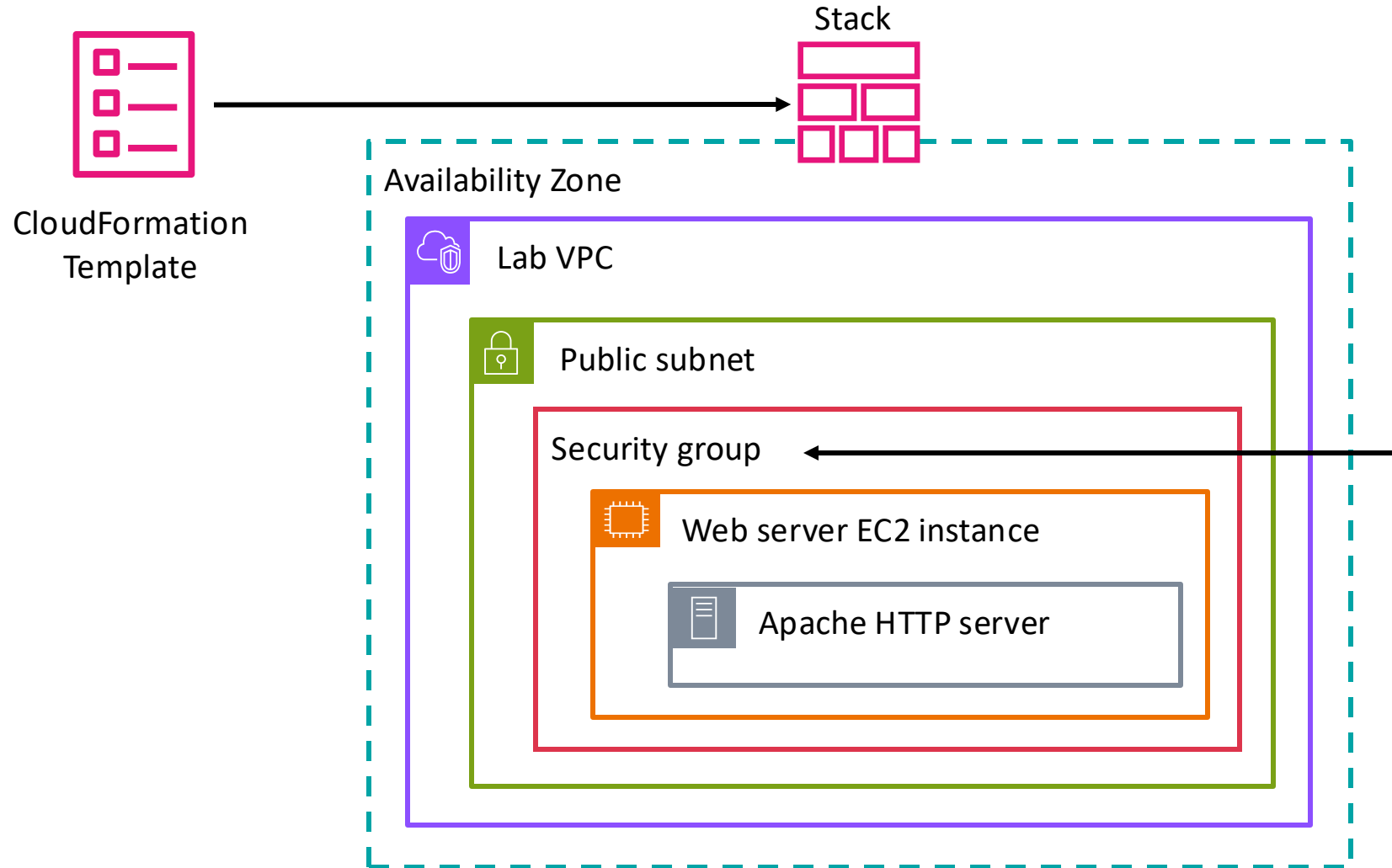
CloudFormation change sets

You can use change sets to **preview changes** before you implement them.



Use the **DeletionPolicy** attribute to preserve or back up a resource when its stack is deleted or updated.

Drift detection



Scenario

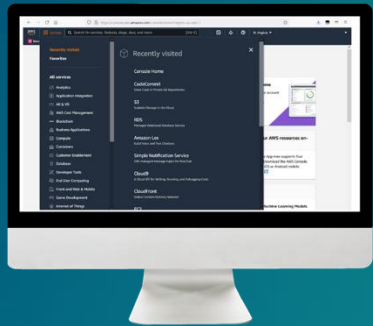
1. An application environment is created by a CloudFormation stack.
2. Later, someone **manually modifies the security group** and opens a new inbound TCP port.
3. Drift detection is run on the stack.
4. All resources except the security group show the result `IN_SYNC`, but the security group shows a status of `MODIFIED` with details.

Scoping and organizing templates

Category	Type of Template
Frontend services	Web interfaces, mobile access, and analytics dashboard
Backend services	Search, payments, reviews, and recommendations
Shared services	Customer relationship management (CRM) databases, common monitoring, alarms, subnets, and security groups
Network	VPCs, internet gateways, virtual private networks (VPNs), and NAT devices
Security	AWS Identity and Access Management (IAM) policies, users, groups, and roles

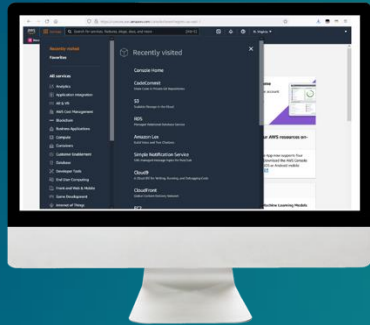
Demo: Analyzing an AWS CloudFormation Template

- This demo uses CloudFormation.
- In this demonstration, you examine a CloudFormation template and analyze the template in depth.



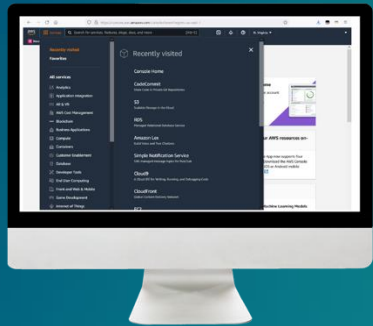
Demo: AWS CloudFormation Resources

- This demo uses CloudFormation and AWS Quick Starts.
- In this demonstration, you are introduced to CloudFormation and the resources available for creating CloudFormation templates.



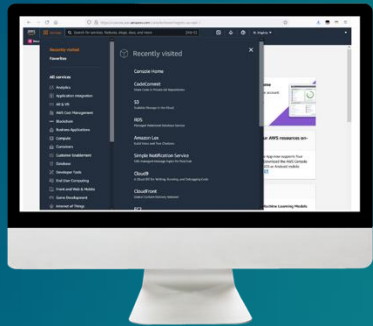
Demo: Reviewing an AWS CloudFormation Template

- This demo uses CloudFormation.
- In this demonstration, you review the changes that you make to a CloudFormation template.



Demo: Using the AWS CloudFormation Console

- This demo uses CloudFormation.
- In this demonstration, you learn how to use the CloudFormation console.



Key takeaways: Customizing with CloudFormation



- CloudFormation is an IaC service that you can use to model, create, and manage a collection of AWS resources.
- CloudFormation IaC is defined in templates that are authored in JSON or YAML.
- A stack is what you create when you use a template to create AWS resources.
- Actions that are available on an existing stack include update stack, detect drift, and delete stack.



Using AWS Quick Starts

Automating Your Architecture

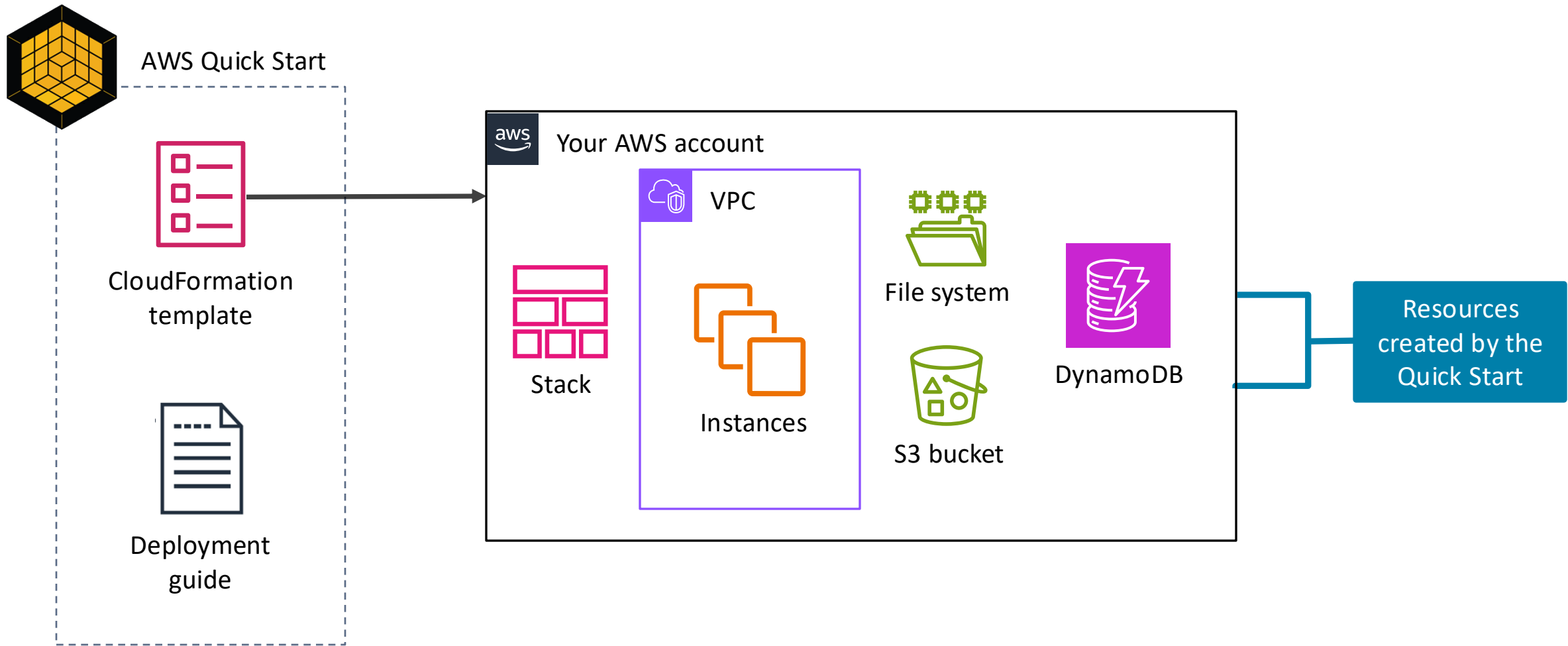
AWS Quick Starts



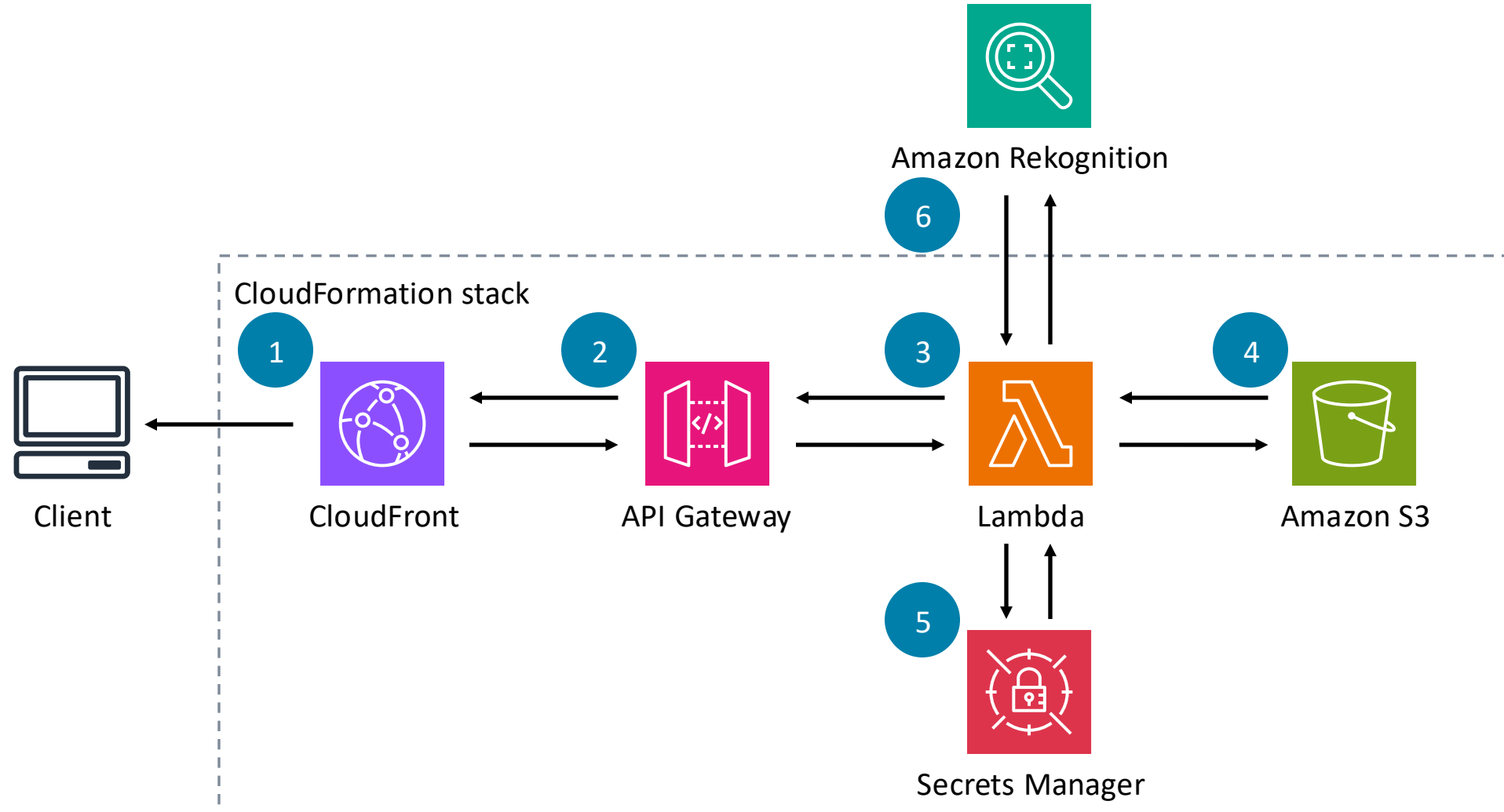
AWS Quick Starts

- Are gold-standard deployments
- Are based on AWS best practices for security and high availability
- Can be used to create entire architectures in less than an hour
- Can be used for experimentation and as the basis for your own architectures

How to use AWS Quick Starts



AWS Quick Starts example



Key takeaways: Using AWS Quick Starts



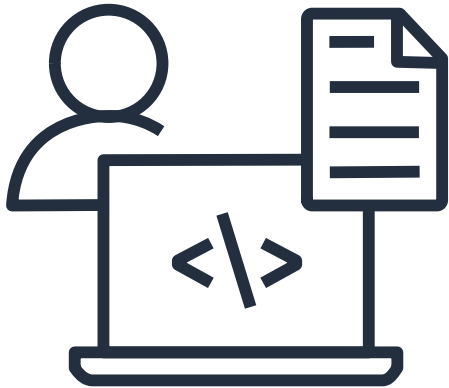
- AWS Quick Starts provide CloudFormation templates built by solutions architects and partners that reflect AWS best practices.
- AWS Quick Starts consists of a CloudFormation template and a deployment guide, which provide details about deployment options and ways to configure the deployment to match your needs.
- AWS Quick Starts can also help you see patterns and practices to accelerate your own template development.



Customizing with Amazon Q Developer

Automating Your Architecture

Challenges of writing infrastructure as code



Human error



Differing skill levels



Size and complexity of
templates



Security vulnerabilities

Amazon Q Developer



Amazon Q Developer

- Generative AI-powered coding assistant
- Designed for developers and IT professionals
- Generates code and helps you understand, build, extend, and operate AWS applications
- Scan code for security vulnerabilities
- Secure and private by design

Amazon Q Developer supports you across the software development lifecycle (SDLC)



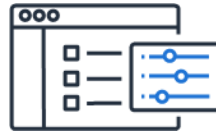
Plan

- Ask questions and get referenceable and contextual guidance.
- Explain code with conversational coding.



Create

- Receive in-line code recommendations for multiple languages.
- Implement features through comments or code prompts.
- Chat in your integrated development environment (IDE).



Test and secure

- Generate unit tests.
- Scan project code for security vulnerabilities and get remediation suggestions.



Operate

- Troubleshoot errors.
- Troubleshoot network connectivity with VPC Reachability Analyzer.



Maintain and modernize

- Modernize code with Amazon Q Developer Agent for code transformation.

Example: Using Amazon Q Developer with CloudFormation

```
example01.yaml x
1  AWSTemplateFormatVersion: '2010-09-09'
2  Description: A sample CloudFormation template
3
4  > ...
13 Resources:
14   # Resource definitions go here
15   # EC2 Instance
16 Outputs:
17   # Output values go here
```

Developer types the start of a desired resource within a YAML file

Amazon Q Developer makes suggestions that the developer can accept

```
example01.yaml x
1  AWSTemplateFormatVersion: '2010-09-09'
2  Description: A sample CloudFormation template
3
4  > ...
13 Resources:
14   # Resource definitions go here
15   # EC2 Instance
```

Suggestion 5 of 5 from Amazon Q

```
EC2Instance:
  Type: AWS::EC2::Instance
  Properties:
    ImageId: XXXXXXXXXXXXXXXXXXXX
    InstanceType: t2.micro
    KeyName: !Ref KeyName
    SecurityGroups:
      - !Ref SecurityGroup
```

The accepted code snippet is added into the template

```
example01.yaml x
1  AWSTemplateFormatVersion: '2010-09-09'
2  Description: A sample CloudFormation template
3
4  > ...
13 Resources:
14   # Resource definitions go here
15   # EC2 Instance
16   EC2Instance:
17     Type: AWS::EC2::Instance
18     Properties:
19       ImageId: XXXXXXXXXXXXXXXXXXXX
20       InstanceType: t2.micro
21       KeyName: !Ref KeyName
22       SecurityGroups:
23         - !Ref SecurityGroup
```

Key takeaways: Customizing with Amazon Q Developer



- Amazon Q Developer is a generative AI coding tool that you can use to generate real-time code suggestions.
- Amazon Q Developer can offer suggestions for writing CloudFormation templates.



Guided lab: Automating Infrastructure Deployment with AWS CloudFormation (CloudFormation lab)

CloudFormation lab tasks:



- In this lab, you perform the following main tasks:
 - Use AWS CloudFormation to deploy a virtual private cloud (VPC) networking layer
 - Use AWS CloudFormation to deploy an application layer that references the networking layer
 - Explore templates with AWS CloudFormation Designer
 - Delete a stack that has a deletion policy
- Open your lab environment to start the lab and find additional details about the tasks that you will perform

Debrief: CloudFormation deployment lab

- What was a best practice that you learned when using CloudFormation to deploy a layered architecture?
- When the CloudFormation stack was created for your VPC, which two outputs did the stack create?



Applying AWS Well-Architected Framework principles to automation

Automating Your Architecture

AWS Well-Architected Framework automation best practices



Operational
Excellence



Security



Reliability



Cost
Optimization

Best practice approach: Prepare – design for operations



Operational
Excellence

Best practices

Perform operations as code.

Make frequent, small,
reversible changes.

Fully automate integration
and deployment.

Best practice approach: Security

Best practices



Security

Automate security best practices.

Best practices approach: Change management – implement change



Reliability

Best practices

Deploy changes
with automation.

Use automation when obtaining or
scaling resources.

Best practices approach: Optimize over time – automating operations

Best practices

Perform automation for operations.



Cost
Optimization

Key takeaways: Applying Well- Architected Framework principles



- Best practices related to automating your architecture include the following:
 - Perform operations as code.
 - Make frequent, small, reversible changes.
 - Fully automate integration and deployment.
 - Automate security best practices.
 - Deploy changes with automation.
 - Use automation when obtaining or scaling resources.
 - Perform automation for operations.



Café lab: Automating Infrastructure Deployment (Café deployment lab)

The evolving café architecture: version 7

Architecture Version	Business reason for update	Technical requirements/ architecture update
V1	Create a static website for a small business.	Host the website on Amazon S3.
V2	Add online ordering.	Deploy the web application and database on Amazon EC2.
V3	Reduce the effort to maintain the database and secure its data.	Separate the web and database layers. Migrate the database to Amazon RDS on a private subnet.
V4	Enhance the security of the web application.	Use Amazon VPC features to configure and secure public and private subnets.
V5	Create separate access mechanisms based on role.	Add IAM groups and attach resource policies to application resources. Add IAM users to groups based on role.
V6	Ensure that the website can handle an expected increase in traffic.	Add a load balancer, implement auto scaling on the EC2 instances, and distribute compute and database instances across two Availability Zones.
V7	Deploy the website in a repeatable way.	
V8	Module 14 info	



Café deployment lab tasks



- In this lab, you will do the following:
 - Deploy a virtual private cloud (VPC) networking layer by using an AWS CloudFormation template
 - Deploy an application layer by using an AWS CloudFormation template
 - Use Git to invoke AWS CodePipeline, and to create or update stacks from templates that are stored in a code repository
 - Use AWS CloudFormation to conditionally build different infrastructures based on environmental variables

Debrief: Café deployment lab

- Where did you store versions of the AWS CloudFormation templates in this lab?
- What did you have to do before you duplicated your application layer template to the Oregon Region?



Module wrap-up

Automating Your Architecture

Module summary

This module prepared you to do the following:

- Recognize when to use architecture automation and why.
- Identify how infrastructure as code (IaC) as a strategy for provisioning and managing cloud resources.
- Identify how to model, create, and manage a collection of AWS resources by using AWS CloudFormation.
- Identify how to use AWS Quick Start CloudFormation templates to set up an architecture.
- Identify uses of Amazon Q Developer.
- Use the AWS Well-Architected Framework principles when designing automation.

Considerations for the cafe

- Discuss how the café lab in this module answered the key questions and decisions presented at the start of this module for the café business.



Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

Consider a situation where you want to create a single AWS CloudFormation template that is capable of creating both a production environment that spans two Availability Zones and a development environment that exists in a single Availability Zone. Which optional section of the CloudFormation template will you want to make use of to configure the logic that will support this?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- A single AWS CloudFormation template
- An environment with two Availability Zones and an environment with a single Availability Zone
- Optional section

Sample exam question: Response choices

Consider a situation where you want to create a single AWS CloudFormation template that is capable of creating both a production environment that spans two Availability Zones, and a development environment that exists in a single Availability Zone. Which optional section of the CloudFormation template will you want to make use of to configure the logic that will support this?

Choice	Response
A	Conditions
B	Outputs
C	Resources
D	Description

Sample exam question: Answer

The answer is A.

Choice	Response
A	Conditions



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.